Power Tracks to Solder Stop

User Manual

power_tracks_to_stop.ulp

An EAGLE / Fusion 360 Electronics user language program that copies the copper of selected signals (tracks and pours) onto the solder stop layers, so that the bare copper is tinned during HASL and the power tracks are reinforced.

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Applies to	EAGLE 9.x, Fusion 360 Electronics (board editor)
Platform	Windows 10
Language of the program	English

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1. Introduction

1.1 Purpose

On a printed circuit board the solder mask normally covers all copper except pads. If a track or a copper pour carries a significant current, it can be reinforced by leaving its copper bare: during the HASL (hot air solder levelling) step a layer of tin is deposited on every exposed copper surface, increasing the cross-section of the conductor.

In EAGLE the solder mask openings are drawn on the stop layers. Opening the mask above a track therefore means drawing, on the stop layer, a shape that covers that track. Doing it by hand (copy the track, paste it on tStop or bStop) is slow and error-prone, especially for pours with many isolation cutouts. **power_tracks_to_stop.ulp** does it automatically for the signals you choose.

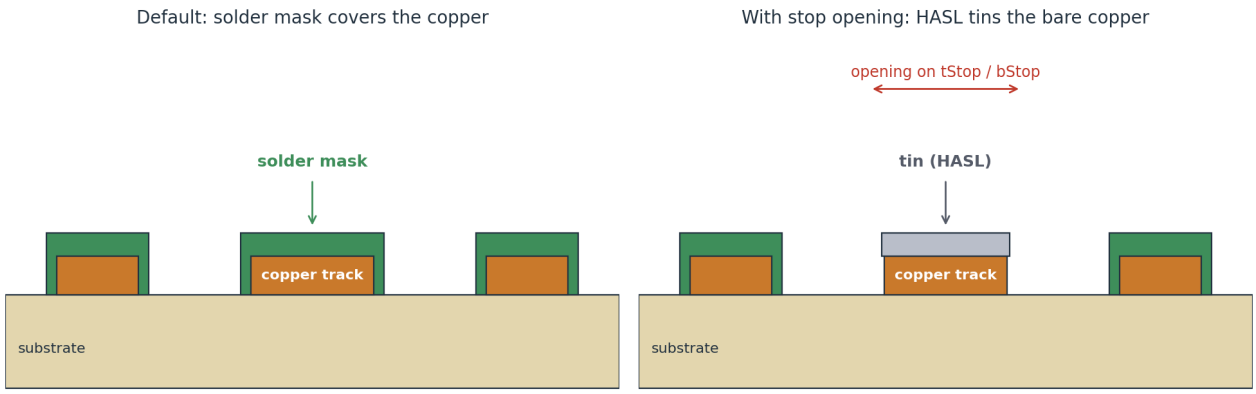


Figure 1. Left: the solder mask covers the copper track. Right: an opening in the stop layer leaves the copper bare, and HASL deposits tin on it.

1.2 Layer mapping

Copper source	Copied to	Layer number
Top copper (wires and pours)	tStop	1 → 29
Bottom copper (wires and pours)	bStop	16 → 30

Only the signals' wires and polygons on layers 1 and 16 are considered. Pads and vias are not copied: the solder stop already opens them.

2. Requirements and installation

Requirements: EAGLE 9.x or Fusion 360 Electronics on Windows 10, with the board editor open on the board to be processed. Pours should be calculated (see section 5.2).

Installation:

- Copy `power_tracks_to_stop.ulp` to a folder of your choice.
- In EAGLE, a file placed in a folder listed under *Options > Directories > User Language Programs* appears in the ULP list. Any other location also works: use the full path in the `RUN` command.

Running it (board editor only). Type in the command line, adapting the path:

```
RUN "C:\Users\you\ulp\power_tracks_to_stop.ulp"
```

In EAGLE you can alternatively use *File > Run ULP...* and browse to the file. If the program is started from the schematic editor or the control panel it stops with a message.

3. Quick start

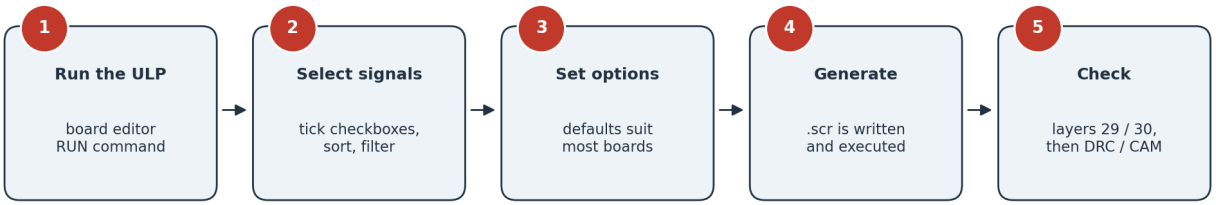


Figure 2. Typical workflow.

1. Save the board. If it contains pours, run RATSNEST so that they are calculated.
2. Run the ULP. The dialog lists every signal that has wires or pours on Top or Bottom copper.
3. Tick the signals to expose. Sorting by *Max w* (descending) brings the widest tracks first; *Width >=* selects them in one click.
4. To include pours, tick *Include polygons (pours)*. The other options can stay at their defaults.
5. Press *Generate*. A summary shows what was created, then the generated script is executed.
6. Check layers 29 and 30 over the copper, then run the DRC and generate your manufacturing files as usual.

> Tip: set the display to show only the stop layer and the copper layer to compare the openings with the copper before sending the board to production.

4. User interface

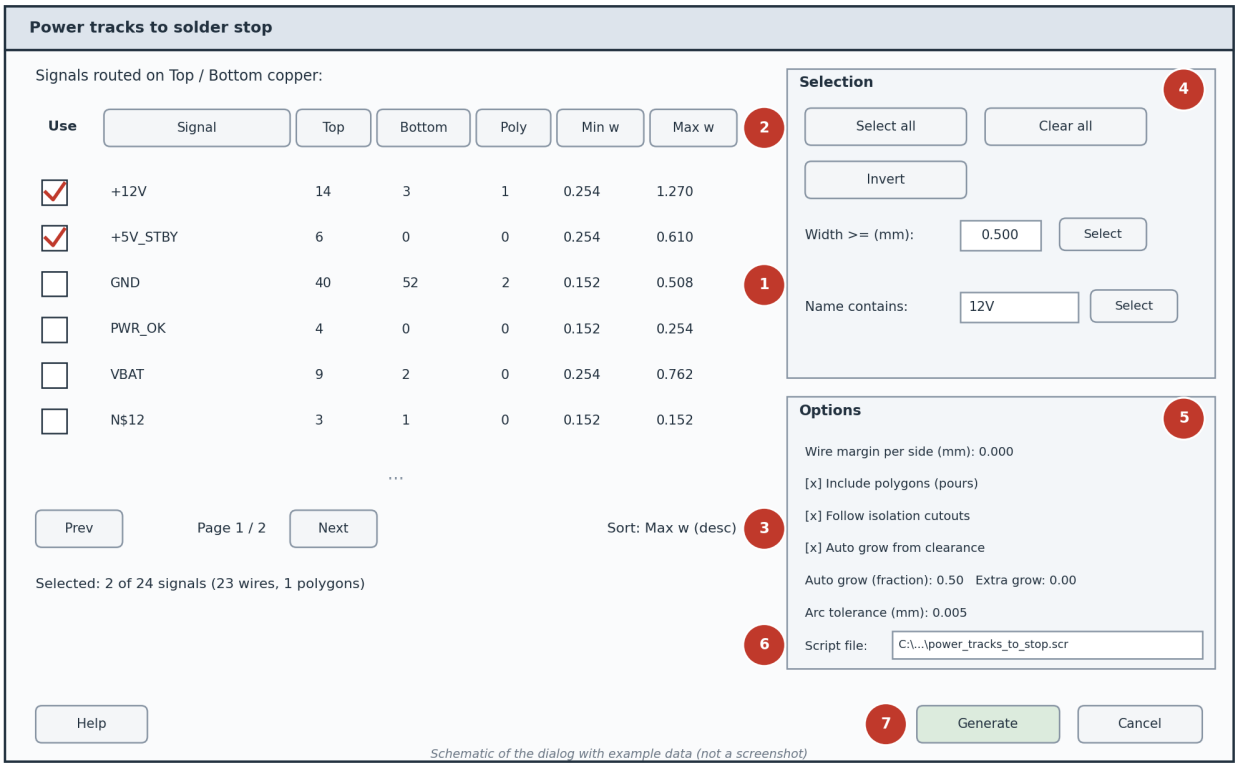


Figure 3. The dialog (schematic, with example data). The numbers refer to table 1.

#	Area	Description
1	Signal list	One row per signal with wires or pours on Top/Bottom. The checkbox on the left marks the signal to expose. 16 rows are shown per page.
2	Column headers	Buttons. Click to sort by Signal, Top, Bottom, Poly, Min w or Max w; click again to reverse the order. Ties are sorted by name.
3	Navigation and status	*Prev* / *Next* change page (ticks are kept). The line on the right shows the current sort. The status line counts selected signals, wires and polygons.
4	Selection tools	Bulk selection, see table 2.
5	Options	Settings of the conversion, see table 3.
6	Script file	Path of the generated .scr file (default: the board's folder).
7	Buttons	*Help* opens a 5-page help. *Generate* writes and runs the script. *Cancel* closes the dialog.

Table 1. Dialog areas.

4.1 Signal list columns

Column	Meaning
Signal	Signal name as in the board.
Top / Bottom	Number of wires of the signal on layer 1 / layer 16.
Poly	Number of polygons (pours) of the signal on Top and Bottom.
Min w / Max w	Smallest and largest wire width on Top/Bottom, in mm. Signals that only have pours show 0.

4.2 Selection tools

Control	Action
Select all / Clear all / Invert	Act on every signal, not only on the visible page.
Width >= (mm) + Select	Adds the signals that have at least one wire as wide as the value (default 0.5 mm).
Name contains + Select	Adds the signals whose name contains the text (case-insensitive).

Table 2. Selection tools. Selections are cumulative; use Clear all to start again.

4.3 Options

Option	Default	Range	Effect
Wire margin per side (mm)	0	0 to 2	Extra width added on each side of every copied wire. It does not apply to pours.
Include polygons (pours)	off	-	Also copies the pours of the selected signals.
Follow isolation cutouts	on	-	Uses the real poured copper, so isolation around other signals stays covered. If off, the polygon outline is used and the cutouts are ignored.
Auto grow from clearance	on	-	Widens the opening around each pour by a fraction of its clearance.
Auto grow (fraction)	0.5	0 to 0.9	Fraction of the clearance used as extra opening per side.
Extra grow per side (mm)	0	0 to 1	Manual extra opening, added to the automatic one.
Arc tolerance (mm)	0.005	0.0005 to 0.1	Maximum distance between a pour arc and the segments approximating it.
Script file	board folder	-	Where `power_tracks_to_stop.scr` is written.

Table 3. Options.

5. How it works

The ULP does not modify the board directly. It writes an EAGLE script (.scr) containing `LAYER`, `WIRE` and `POLYGON` commands and then executes it with `SCRIPT`.

5.1 Wires

Each wire of a selected signal on layer 1 or 16 is copied to layer 29 or 30 with the same end points, the same curvature (arcs are preserved) and the same width, plus twice the wire margin. Before writing a wire the ULP checks whether an identical wire already exists on the stop layer; if so it is skipped and counted as a duplicate, so the wire part can be generated again safely.

5.2 Pours (polygons)

A pour cannot simply be copied: its outline does not show the isolation around other signals. The ULP reads the **poured contour** that EAGLE computed, which includes those cutouts. This is why the pour must be calculated (`RATSNEST`). The steps are:

1. The contour wires are chained into closed loops. Arcs are replaced by short segments, controlled by the arc tolerance.
2. Each loop is classified by nesting: an outer boundary, a hole (isolation around a pad or another signal), or an island inside a hole.
3. Every hole is joined to its outer boundary by a bridge of zero width, producing a single polygon. The bridge is chosen so that it lies entirely inside the copper.
4. That polygon is written on the stop layer with a hairline edge: it only fills the area.
5. The border of the pour is drawn separately with wires as wide as the pour width (plus the grow), following the same contour. This reproduces the copper edge, which extends half the pour width beyond the contour line.

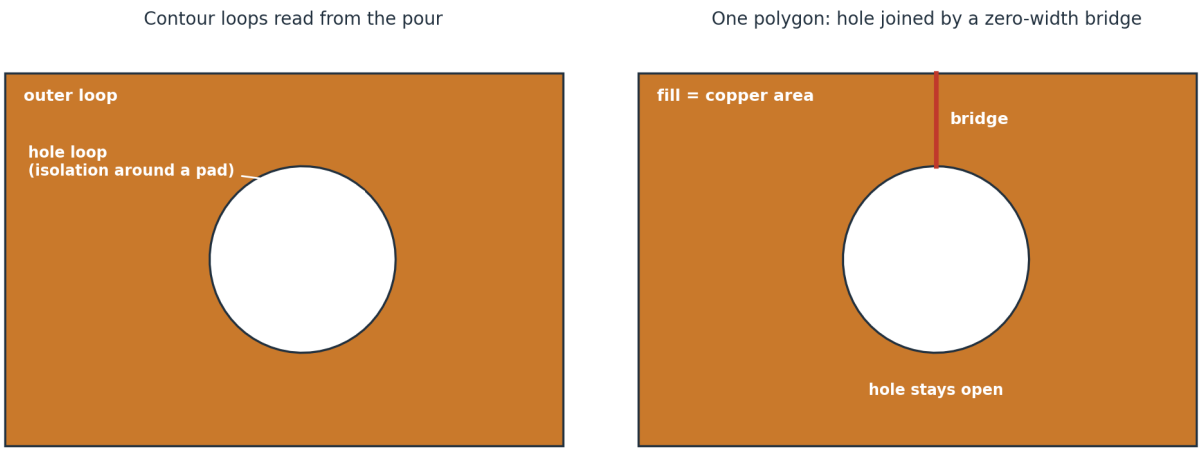


Figure 4. Left: loops read from the pour. Right: the hole is joined to the outer boundary by a zero-width bridge (drawn thick here for visibility) so that one polygon describes the whole copper area.

If a pour is not calculated, or *Follow isolation cutouts* is off, the polygon outline is used instead. The summary reports the polygons that fell back to the outline because they were not calculated.

5.3 Border width and automatic grow

EAGLE draws the pour border as a wire of the pour width centred on the contour. The real copper edge is therefore half of that width beyond the contour. The stop border wire has the pour width plus twice the *grow*, so the opening reaches the copper edge and then *grow* beyond it.

The automatic grow is **fraction × clearance**, where the clearance is the pour's isolate value or, if it is 0, the clearance of the signal's net class (0.2 mm if both are 0). With the default fraction of 0.5 the opening ends halfway into the isolation gap, which guarantees that it never reaches the neighbouring copper. The extra opening compensates small differences at the copper edge and makes sure the copper is fully exposed.

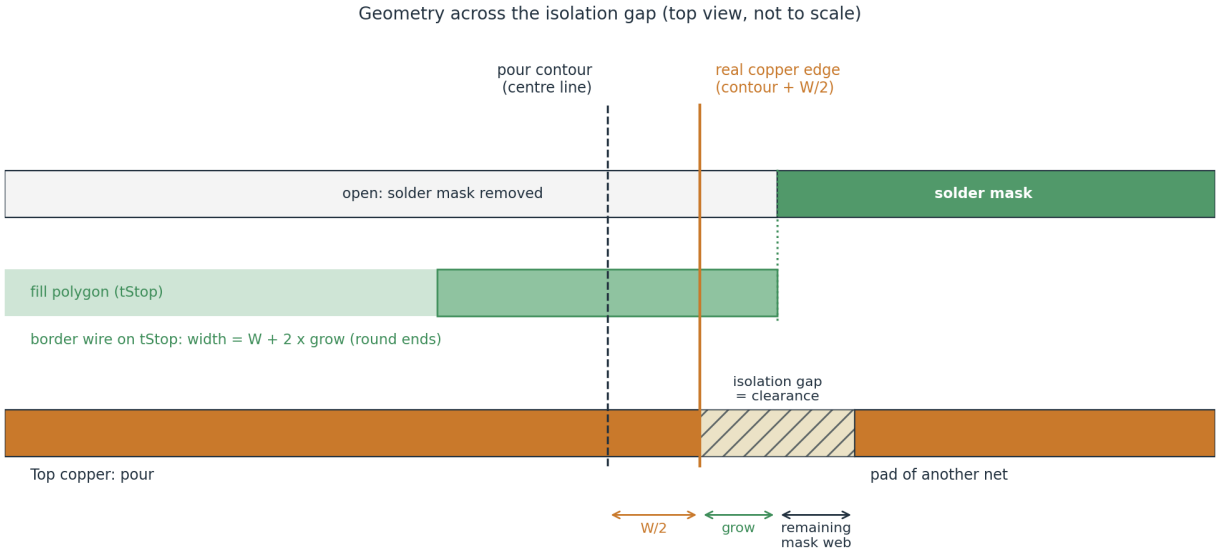


Figure 5. Geometry across the isolation gap between a pour and the pad of another net.

5.4 Generated script

The script is plain text and can be opened with any editor. Example (values are illustrative):

```
GRID mm;
LAYER 29;
```

```
WIRE 0.5080 (12.7000 25.4000) (30.4800 25.4000);
LAYER 29;
POLYGON 0.0010
  (10.0000 10.0000) (50.0000 10.0000) (50.0000 40.0000) (10.0000 40.0000)
  (10.0000 10.0000);
LAYER 29;
WIRE 0.6556 (10.0000 10.0000) (50.0000 10.0000);
```

6. Verification and good practice

- **Save the board before generating.** To revert, use undo or reload the saved file; a long script can produce many undo steps.
- **Mask web.** The opening is deliberately a little wider than the copper. Check with your manufacturer the minimum solder mask web between exposed copper and neighbouring pads and lower the auto grow if needed.
- **Calculate pours first** (RATSNEST) and regenerate after any change to the pours or to nearby copper.
- **Do not generate the same pours twice** without removing the previous stop polygons: duplicate wires are detected, polygons are not.
- **Inspect the result** with only layers 1/16 and 29/30 visible, and look at the Gerber output before ordering.
- **Large pours** contain many vertices; processing takes a few seconds.

7. Limitations

- Only wires and polygons that belong to signals are listed. Wires or polygons that have no signal are ignored.
- Pads, vias, holes and text are not copied.
- The opening around a pour follows the contour and the auto grow rule; it is not a recalculation of the pour.
- The default auto grow fraction (0.5) is a starting point; it has been chosen for safety, not derived from a specific manufacturer's rules.
- In Fusion 360 Electronics the board may be stored in the cloud: if the script file cannot be written, choose a local folder in *Script file*.

8. Troubleshooting

Symptom	Likely cause	What to do
Message: run from the board editor	The ULP was started from another editor.	Open the board and run it there.
A signal with only a pour is missing	The pour is not on layer 1 or 16, or the signal has no wires or pours on Top/Bottom.	Check the layer of the pour.
Summary warns about uncalculated polygons	The pour was not calculated, so only the outline was used.	Run RATSNEST and generate again.
Isolation cutouts are covered by the opening	*Follow isolation cutouts* is off, or the outline fallback was used.	Turn the option on and calculate the pours.
Thin strip of copper is still covered by mask	The grow is too small for the copper edge.	Increase the auto grow fraction or the extra grow per side.

Symptom	Likely cause	What to do
Opening reaches neighbouring pads	The grow is too large for the clearance.	Lower the auto grow fraction and the extra grow.
Generated polygons or wires appear twice	The script was generated twice.	Undo or delete the stop objects and generate once.
Script file cannot be written	The board folder is read-only or cloud-only.	Change *Script file* to a local folder.
Compile or runtime error	-	Report the exact message and the line number.

Appendix A. Commands in the generated script

Command	Used for
GRID mm;	Sets millimetres as the unit for the coordinates that follow.
LAYER 29; / LAYER 30;	Selects tStop or bStop.
WIRE w (x1 y1) (x2 y2);	Copied wire, or border wire of a pour. An arc is written with its curvature between the two points.
POLYGON w (x y) ... ;	Fill polygon of a pour, with a hairline edge width w.